



OPEN Energy absorption and rebound behavior of 3D-printed TPU lattice structures

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Unmanned aerial vehicles (UAVs) often encounter complex ground conditions during takeoff and landing, especially on unpaved surfaces such as snow, sand, grass, and slopes, as well as under extreme conditions like strong winds and dense fog. In these scenarios, the landing gear may experience significant impact loads and vibrations. If the structure fails to absorb energy effectively and rebound promptly, it may compromise airframe stability, onboard system reliability, and operational safety. To optimize the energy absorption–rebound performance of UAV landing gear, this study employed thermoplastic polyurethane (TPU 98 A) to design five homogeneous and gradient lattice structures with varying cell parameters, fabricated via fused deposition modeling (FDM). Quasi-static compression tests were conducted to investigate load–displacement behavior, energy absorption, and rebound performance. A multidimensional evaluation system was established based on total energy absorption (EA), specific energy absorption (SEA), hysteresis loop area (HLA), energy recovery ratio (ERR), residual displacement ratio (RDR), and stiffness degradation ratio (SDR), enabling an integrated assessment of energy absorption and rebound. Results demonstrate that TPU lattice structures achieve high energy absorption while maintaining excellent rebound performance and structural stability.

Keywords Thermoplastic polyurethane (TPU), Quasi-static compression, Energy absorption, Vibration damping

Polymeric porous materials have attracted considerable attention for vibration-damping and energy-absorption applications in aerospace, transportation, and precision engineering due to their high specific strength, tunable design flexibility, and favorable damping behavior^{1–6}. Among these materials, thermoplastic polyurethane (TPU) combines elasticity and toughness, exhibiting remarkable energy-dissipation capability. Compared with conventional metallic honeycomb structures, 3D-printed TPU lattices and honeycombs are lighter, geometrically customizable, and particularly suited for unmanned aerial-vehicle (UAV) vibration-isolation components^{7–11}.

Recent advances in lattice design has focused on optimizing mechanical anisotropy and hierarchical control. With the development of additive manufacturing, researchers have designed polymeric and composite lattices with controllable anisotropy, gradient distribution, and hierarchical geometry^{12–15}. Gharehbaghi et al.¹⁶ proposed a new hyperbolic-beam-based hexagonal lattice model and validated its mechanical behavior through analytical, numerical, and experimental methods, confirming its suitability for porous materials. Subsequent studies by Gharehbaghi and collaborators^{17–19} investigated continuous glass-fiber-reinforced PLA honeycombs and lozenge-grid architectures under various loading directions, revealing enhanced stiffness-to-weight ratios and directional energy absorption. Avarzamini et al.¹⁹ further demonstrated that functionally graded, fiber-reinforced sinusoidal lattices exhibit superior orientation-dependent energy absorption. Collectively, these studies emphasize that combining fiber reinforcement with graded topology substantially improves crashworthiness and mechanical efficiency.

Similarly, Sadeghzade et al.²⁰ examined 3D-printed octagonal bipyramid gradient lattices and found that material choice and cell gradation critically affect energy-absorption efficiency. Ghorbani et al.²¹ designed a bio-inspired glass-fiber-reinforced cellular structure, employing particle-swarm optimization to enhance stiffness and impact resistance. Gharehbaghi and Farrokhabadi²² optimized a cruciform composite lattice through local-degradation modeling combined with an innovative print-path strategy, achieving high energy-absorption efficiency and excellent crash tolerance. Collectively, these studies confirm that hybrid designs integrating geometric grading, reinforcement, and tailored deformation mechanisms outperform homogeneous lattices.

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At the macroscopic scale, hierarchical organization can markedly improve structural behavior, enabling performance customization for lightweight designs^{23–29}. Functionally graded and hierarchical cellular architectures offer reduced density while improving stiffness, strength, and impact resistance^{30–34}. Bates et al.⁸ fabricated superelastic TPU honeycombs with graded densities via fused-filament fabrication (FFF), achieving stable cyclic compression. Beloshenko et al.⁹ and Yang et al.¹⁰ further confirmed that optimized cell topology enhances stiffness, ductility, and rebound efficiency in flexible TPU lattices. Recent reviews^{35–42} also indicate that gradient or hierarchical honeycombs enable synergistic energy absorption and recovery, highlighting their potential in adaptive damping systems.

Compared with metallic and fiber-reinforced lattice benchmarks^{13,37}, the proposed TPU structures offer lower density, ease of fabrication, and superior elastic recoverability, offering distinct advantages for UAV support applications. While previous single-gradient or homogeneous topologies enhanced energy absorption only along specific orientations^{8,37}, the present dual-gradient design achieves balanced stiffness retention and rebound efficiency through interlayer reinforcement. This combination of material flexibility and geometric gradation provides a lightweight and manufacturable solution for UAV support structures requiring effective vibration attenuation^{8,9}.

In summary, TPU-based lattice structures have emerged as promising candidates for high-performance vibration-isolation and energy-absorbing applications due to their outstanding designability and manufacturability. In this study, five novel lattice configurations were fabricated, emphasizing simplicity and stability, and incorporating a bottom-beam element to improve global stiffness. To satisfy the integrated demands of light weight, vibration damping, and structural support for UAV landing systems, hybrid gradient and symmetric-gradient arrangements with inter-layer reinforcement are proposed. These designs enable simultaneous enhancement of energy-absorption efficiency and rebound stability. Subsequent quasi-static compression experiments systematically evaluate their mechanical responses and integrated energy-absorption–rebound synergy performance.

Test specimen Preparation and static compression testing

Figure 1 shows the workflow for TPU fabrication and performance testing. It can be divided into three steps: test specimen manufacturing, experimentation, and comprehensive performance evaluation. Fig. 2 shows 3D models of the five lattice structure and their application scenarios.

Step 1 involves designing five lattice test specimens with varying unit cell parameters using 98 A-TPU filament and an FDM 3D printer. Step 2 shows the compression testing according to GB/T 1041–2008 standard (Plastics-determination of compressive properties). Step 3 establishes a multidimensional performance evaluation framework and an integrated energy absorption–rebound assessment method. The test specimens' energy absorption and rebound characteristics were evaluated to optimize the structural design.

Model structure design

To meet the demand for lightweight buffering structures for UAVs under landing impact and in-flight vibration environments, TPU lattice structures are used as vibration-damping and energy-absorbing components for the UAV landing gear. Their topological parameters and deformation mechanisms were systematically analyzed. Figure 2(g) shows the application scenario, where the yellow-highlighted region represents the 3D-printed landing gear structure. The five lattice topology designs proposed in this work were installed at this location.

The three-dimensional models of the five lattice structures comprise arrays of regular hexagonal cells with varying unit cell parameters in Fig. 2. The top surface is the upper face, with the cells connected to the first layer. The subsequent layers are arranged sequentially along the negative Z-axis. Figure 2(c) shows a graded lattice structure. Figure 2(d) shows a gradient structure symmetric along the central axis. The models in Fig. 2(a)–(e)

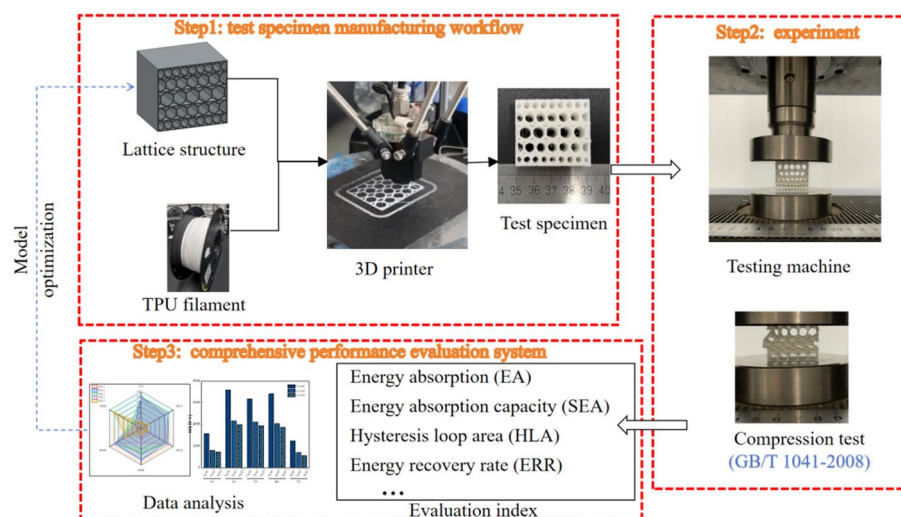


Fig. 1. TPU material structure fabrication and performance testing.

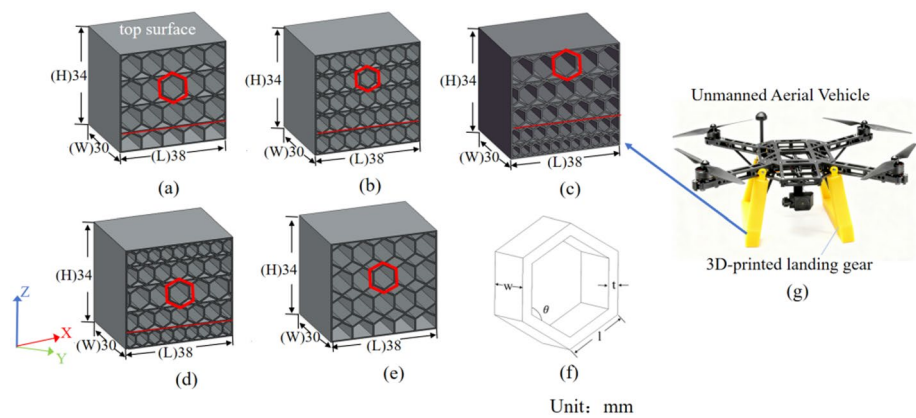


Fig. 2. 3D models of the five lattice structures and their application in UAV landing gear: (a)-(e) Lattice structures, (f) unit cell geometry, (g) unmanned aerial vehicle.

ID	Region layer	l (mm)	t (mm)	w (mm)	θ (°)	Beam structure
1#	Homogeneous	4.020	0.5	30	120	Yes
2#	Homogeneous	2.885	0.5	30	120	Yes
3#	1	4.040	0.5	30	120	Yes
	2	3.465	0.5	30	120	
	3	2.885	0.5	30	120	
	4	2.310	0.5	30	120	
	5	1.730	0.5	30	120	
4#	1 and 5	1.125	0.5	30	120	Yes
	2 and 4	1.545	0.5	30	120	
	3	1.875	0.5	30	120	
5#	Homogeneous	4.020	0.5	30	120	No

Table 1. The unit cell parameters for models 1# to 5#.

Nozzle temperature (°C)	Nozzle diameter (mm)	Layer height (mm)	Perimeter(mm/s)	Infill (mm/s)	Travel (mm/s)	Fill pattern	Top/bottom layers
220	0.4	0.2	80	40	120	Line	3

Table 2. The FDM 3D printing parameters.

are labeled as 1# to 5#, respectively. All models adhere to exact dimensions with a length (L) of 38 mm, a width (W) of 30 mm, and a height (H) of 34 mm. The red hexagons indicate unit cell structures, and red lines represent horizontal beam structures. Models 1# to 4# include horizontal beams between layers. Figure 2f illustrates the unit cell configuration, indicating the wall thickness (t), side length (l), width (w), and hexagonal interior angle (θ) (Table 1).

The side lengths of the regular hexagonal unit cells for the homogeneous structures 1#, 2#, and 5# are 4.02 mm, 2.885 mm, and 4.02 mm, respectively. In the gradient structure 3#, the side lengths of the hexagonal unit cells in each layer are 4.04 mm, 3.465 mm, 2.885 mm, 2.310 mm, and 1.730 mm, arranged in a descending order. For the centrally symmetric gradient structure 4#, the side lengths of the regular hexagonal unit cells are 1.125 mm, 1.545 mm, 1.875 mm, 1.545 mm, and 1.125 mm, exhibiting symmetry about the central axis.

Test specimen preparation

TPU 98 A was procured from Ruiben Information Technology Co., Ltd., Guangdong, China. The specimens were fabricated using a fused deposition modeling (FDM) printer (CoLiDo D315 Plus, Print-Rite, Guangdong, China). The FDM3D printing parameters are listed in Table 2.

Key printing settings include nozzle temperature of 220 °C, nozzle diameter of 0.4 mm, Perimeter speed of 80 mm/s, Infill speed of 40 mm/s, Travel speed of 120 mm/s, layer height of 0.2 mm, linear infill pattern, and the Top or Bottom layers of 3. Figure 3 shows FDM 3d printing of the five TPU test specimens.

The masses of specimens 1# to 5# were 12.75 g, 16.67 g, 14.62 g, 15.79 g, and 10.12 g, respectively.

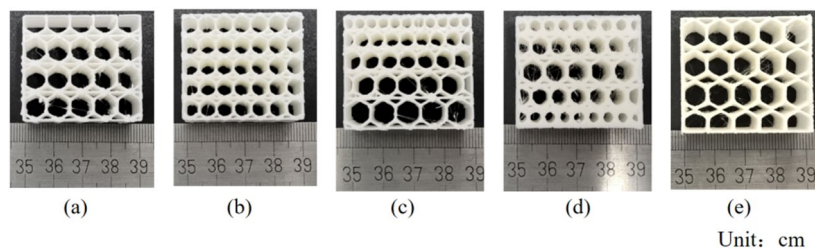


Fig. 3. The five TPU test specimens (a) to (e) show fabricated test specimens 1# to 5#, respectively.



Fig. 4. Quasi-static compression of TPU specimens at displacements (0 and 20 mm): (a)–(e) specimens 1# to 5#.

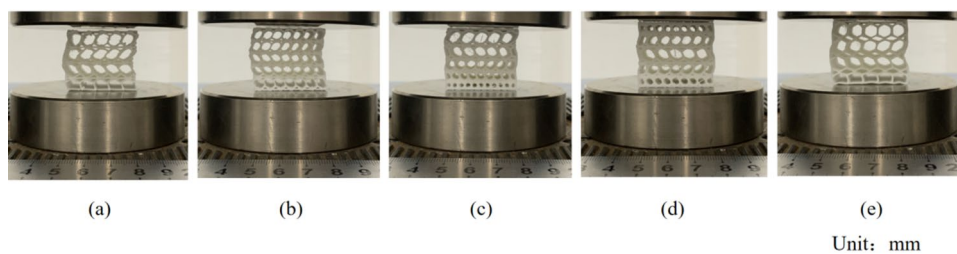


Fig. 5. Post-unloading morphology of specimens 1# to 5# after the first cycle: (a) to (e) are specimens 1# to 5#, respectively.

Quasi-static compression test

Quasi-static compression tests were conducted following the GB/T 1041–2008 standard (plastics-determination of compressive properties). The experiments were performed using a universal testing machine (AUTOGRAPH AG-X, Shimadzu, Japan) at a room temperature of 25 °C. A three-cycle test was conducted at a constant loading speed of 5 mm/min within a displacement range of 0 to 20 mm. Figure 4 presents the morphological evolution of specimens 1# to 5# at displacements of 0 mm, 5 mm, 10 mm, 15 mm, and 20 mm, respectively. The deformation process includes the elastic region, the plateau (stress) region, and the densification stage. Figure 5 presents post-unloading morphology of five TPU specimens after the first cycle.

Numerical simulation method for quasi-static experimental compression

To investigate the deformation mechanisms of the five TPU lattice structures during quasi-static compression, numerical simulations were performed using finite element simulation (ANSYS) to the first compression cycle. The materials parameters were derived from 98 A TPU tensile data provided by Supplier Network⁴³, with an equivalent Young's modulus of approximately 200 MPa. The Poisson's ratio was set to 0.46 according to the experimental data from BASF Polyurethanes GmbH⁴⁴, and the density was set to 1.20 g/cm³. The lattice models were discretized using SOLID187 tetrahedral elements, with an overall mesh size of 0.8 mm. The bottom surface of each model was fully constrained, while a 20 mm displacement was applied to the upper platen to reproduce the experimental loading conditions. Figure 6 presents the numerical simulation results of specimens 1# to 5# at a compression displacement of 20 mm. Figure 6(a)–(c) shows the Morphology and load–displacement comparison between experiments and simulations for TPU specimens 1# to 5#, respectively.

The finite element simulations effectively captured the dominant deformation modes of the five TPU lattice structures during compression. For specimen 1#, the simulation showed layer-by-layer buckling from top to bottom, accompanied by noticeable bending and local folding of the cell walls, consistent with the hierarchical compression characteristics of its uniform cell size. Specimen 2#, with a denser cell configuration, exhibited early global densification and pronounced vertical shear buckling in the simulation, indicating the formation of a large compressed zone at the initial stage of loading. The gradient lattice of specimen 3# showed progressive collapse from large cells to small cells, with the simulated collapse sequence closely following the structural gradient, demonstrating the influence of topological grading on the compression path. Specimen 4#, featuring a symmetric gradient, displayed symmetric folding and densification around the middle region, reflecting the concentrated deformation induced by its symmetric cell distribution. Previous studies have shown that lattice topology and gradient symmetry critically affect collapse propagation paths under compression⁴⁵, consistent with the distinct collapse behaviors observed in specimens 3# and 4#. The structure of specimen 5#, which lacks transverse beams, exhibited diagonal shear collapse in the simulation, with an irregular global failure direction, indicating that the absence of reinforcing beams significantly reduced structural stability.

Overall, the simulations reveal the key deformation mechanisms during the first compression stage, including buckling initiation, folding propagation paths, and overall collapse modes, all of which agree closely with the experimental observations.

Testing evaluation method

Conventional evaluations of lattice structures mainly rely on single performance metrics, such as specific energy absorption (SEA) or energy recovery ratio (ERR) to characterize deformation efficiency or resilience. However, these isolated indicators fail to capture the interdependent behavior among energy dissipation, elastic recovery, and structural degradation under cyclic compression. To address this limitation, the present study proposes a multidimensional evaluation system integrating six normalized indicators: EA, SEA, HLA, ERR, RDR, and SDR. This approach provides a comprehensive assessment of both energy absorption capacity and rebound sustainability. By visualizing the overall performance of different lattices using radar charts and quantifying their cyclic mechanical stability, this data-driven method overcomes the reductionist limitation of single-metric evaluations and provides a systematic tool for the selection and optimization of TPU lattices for UAV vibration-damping applications.

Energy absorption performance evaluation

Energy absorption performance was evaluated using the EA, SEA, and HLA. EA represents the total energy absorbed by the structure during the loading process, indicating its impact resistance, and is calculated as:

$$EA = \int_0^{s_{\max}} F(s) ds \quad (1)$$

where EA is the total absorbed energy, s is the displacement, and F is the applied load.

SEA is used to calculate the energy absorbed per unit mass of each test specimen. This can be expressed as follows:

$$SEA = \frac{EA}{m} = \frac{\int_0^s F ds}{m} \quad (2)$$

where SEA is the specific energy absorbed by the structure, and m is the mass of the TPU specimen.

HLA quantifies the energy dissipated during the loading-unloading cycles, demonstrating the structure's cushioning and damping capability. This can be expressed by the following relationship:

$$A_{\text{hysteresis}} = \oint F \approx \sum_{i=1}^{n-1} \frac{(F_i + F_{i+1})}{2} \cdot (s_{i+1} - s_i) \quad (3)$$

where $A_{\text{hysteresis}}$ is the area enclosed by the hysteresis loop and F_i, s_i is the load and displacement at the i_{th} data point. The absolute value of the loop is calculated in a clockwise direction.

Rebound performance evaluation

Rebound performance was assessed using ERR, RDR, and SDR. The ERR measures the ratio of unloading to loading energy given by the following expression:

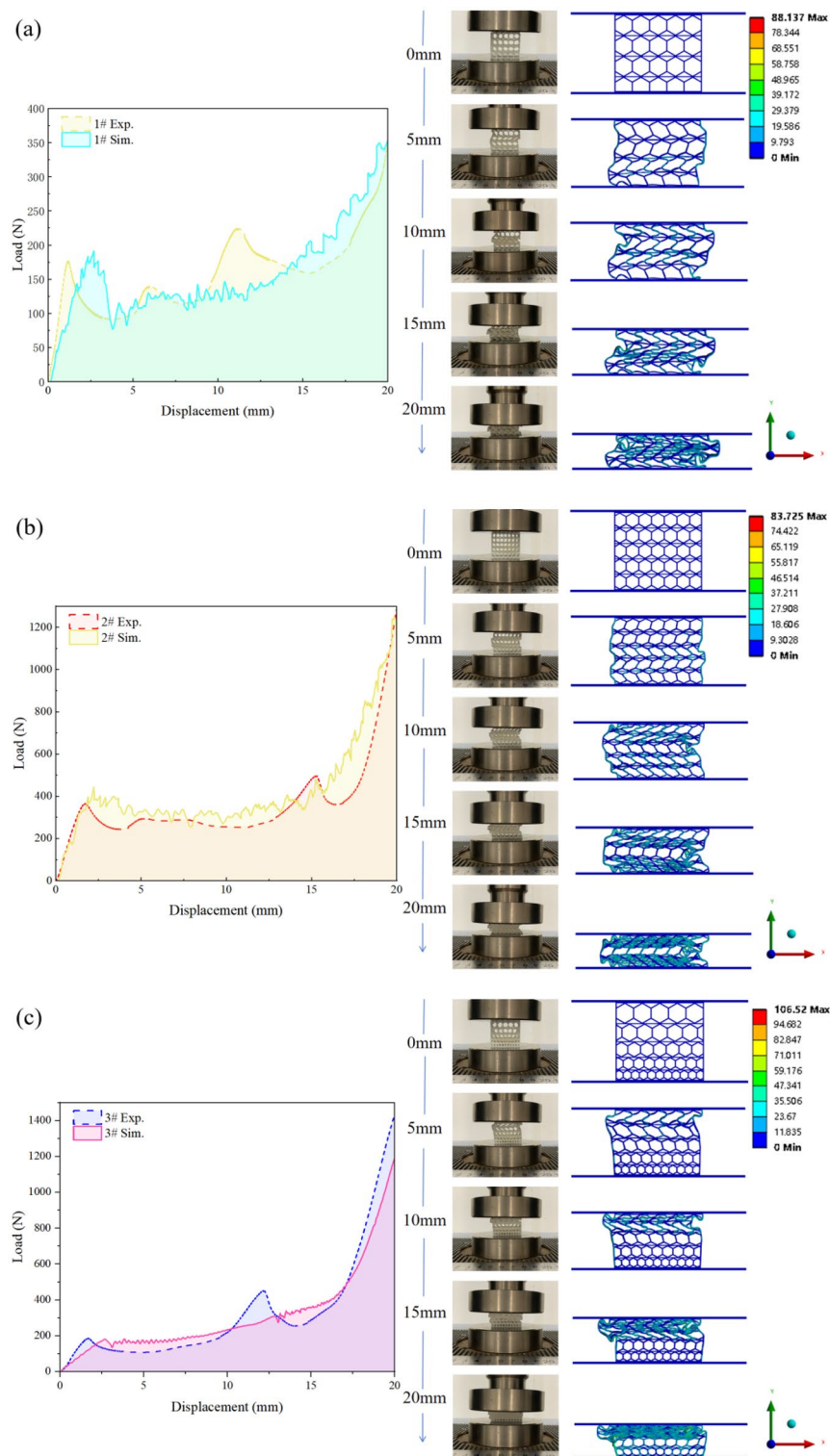


Fig. 6. The simulated and experimental deformation morphologies of specimens 1#–5# at a compression displacement of 20 mm: (a)–(e) specimens 1#–5#.

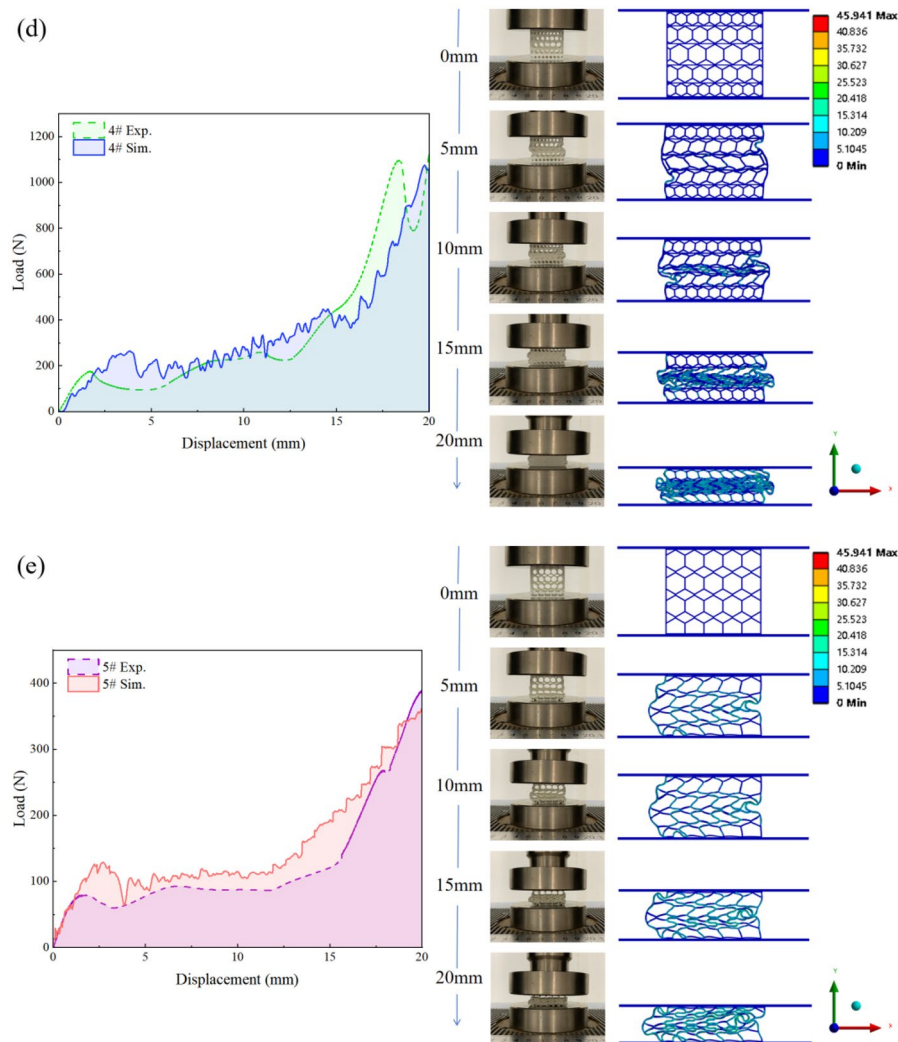


Fig. 6. (continued)

$$ERR = \frac{E_r}{EA} \times 100\% \quad (4)$$

where ERR is the energy recovery ratio, and E_r is the energy released during unloading.

RDR quantifies the extent of permanent deformation after unloading, which can be expressed as follows:

$$RDR = \frac{h_r}{h_0} \times 100\% \quad (5)$$

where RDR is the residual displacement ratio, h_r is the residual height after unloading, and h_0 is the initial height of the specimen.

SDR quantifies a decrease in structural stiffness over cyclic loading, which is expressed as follows:

$$\left\{ \begin{array}{l} K_i = \frac{F_{max,i} - F_{min,i}}{x_{max,i} - x_{min,i}} \times 100\% \\ SDR = \frac{K_3}{K_1} \times 100\% \end{array} \right. \quad (6)$$

where K_i is the secant stiffness in the i^{th} cycle. $F_{max,i}$ and $F_{min,i}$ are the peak and minimum loads, respectively. $x_{max,i}$ and $x_{min,i}$ are corresponding displacements to peak and minimum loads, respectively. SDR is the stiffness retention from the first to the third cycle.

Comprehensive evaluation of energy absorption–rebound synergy

For positive indicators (EA , SEA , HLA , ERR , and SDR), normalization was performed using the following formula:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (7)$$

where X is the original metric value, X' is the normalized dimensionless value (ranging from 0 to 1), $X_{\min}$ is the minimum value of the metric before normalization, and $X_{\max}$ is the maximum value of the metric before normalization.

For negative indicators (RDR), normalization was performed using the following formula:

$$X' = 1 - \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (8)$$

After normalization, the area of the hexagonal radar chart was calculated.

$$\left\{ \begin{array}{l} A = \frac{1}{2} \left| \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i) \right| \\ x_i = r_i \cos \theta_i \\ y_i = r_i \sin \theta_i \end{array} \right. \quad (9)$$

where A is the area of the radar chart polygon, n is the number of performance dimensions (taken as 6), r_i is the normalized value of the i^{th} indicator. x^i and y^i are the 2D coordinates of each performance point. θ_i is the angular position of each indicator on the radar plot. The area is computed using the vector cross product between adjacent points.

Results and analysis

Experimental results and analysis

Figure 7 shows the load-displacement curves for compression loading, unloading, and hysteresis loops of specimens 1# to 5# across three cyclic tests. These curves reveal the energy absorption, rebound performance, and stiffness degradation characteristics of the TPU lattice structures. In the plots, the color coding is as follows: black for specimen 1#, red for 2#, blue for 3#, green for 4#, and purple for 5#.

Figure 7(a)–(c) shows the loading responses. In Fig. 7(a), specimens 2# and 4# exhibit higher load responses and compressive strength, while 5# shows the lowest load-bearing capacity and stiffness. In Fig. 7(b) and 7(c), the 2# curve steeply rises in the first 5 mm of displacement, indicating early load-bearing. Figures 7(d)–7(f) are the unloading paths. Specimens 2# and 3# show rebound curves closely following their loading paths, indicating strong elastic recovery and high energy return efficiency. In contrast, 5# shows a significant offset between loading and unloading, indicating high residual deformation and poor rebound dominated by plastic dissipation. Figures 7(g) to 7(i) illustrate the evolution of hysteresis loop areas across three cycles. 2# and 4# consistently larger loop areas indicating sustained energy dissipation, while 5# shows the smallest and decreasing hysteresis area, suggesting a diminishing damping capacity.

Single-index results and analysis

(1) Single-Index Results.

Table 3 lists the evaluated performance metrics (EA, SEA, HLA, ERR, RDR, and SDR) for specimens 1# to 5# across three consecutive compression cycles.

In terms of energy absorption performance, specimen 1# showed $EA = 3142.801 \times 10^{-3}$ J, $SEA = 246.494 \times 10^{-3}$ J/g, $HLA = 2445.521 \times 10^{-3}$ J during cycle 1; $EA = 1611.929 \times 10^{-3}$ J, $SEA = 126.426 \times 10^{-3}$ J/g, $HLA = 948.985 \times 10^{-3}$ J during cycle 2; and $EA = 1454.382 \times 10^{-3}$ J, $SEA = 114.069 \times 10^{-3}$ J/g, $HLA = 819.913 \times 10^{-3}$ J during cycle 3. In terms of resilience performance, the ERR are 24.347%, 43.832%, and 46.482%; the RDR are 26.460%, 30.031%, and 31.726%. After three cycles, the SDR is 111.254%.

(2) Results and Analysis.

Figure 8 compares the energy absorption and rebound performance across the three cycles.

In Fig. 8(a), specimen 2# exhibits the highest total energy absorption (EA) in all cycles, with 3979.978×10^{-3} J in cycle 3, significantly outperforming the other cycles. This superior performance is attributed to its dense pore structure and short cell walls, which enable early and extensive buckling (Fig. 4). Specimen 4# ranks second, while 5# performs worst (1124.577×10^{-3} J) due to disordered collapse and fewer deforming units.

Figure 8(b) shows that specimen 3# achieves the maximum SEA in cycle 3, reaching 264.621×10^{-3} J/g. This occurs due to its gradient pore architecture, which facilitates layer-by-layer progressive collapse. Although 2# and 4# exhibit high EA, their heavier mass decreases SEA. Despite being the lightest, 5# also exhibits the lowest SEA, indicating that low weight alone is not responsible for high energy efficiency. These results demonstrate that the relationship among pore size, mass density, and SEA is inherently nonlinear. This is because variations in geometric scale and topology modify collapse modes, plateau stress evolution, and densification behavior, rather than producing a proportional density-SEA scaling⁴⁶.

In Fig. 8(c), the HLA is greatest for 2# and 4# in cycle 1 (4820.564×10^{-3} J and 4948.120×10^{-3} J, respectively), showing broad plastic buckling zones and strong energy dissipation. 3# exhibits stable energy absorption, while 5# reveals the lowest energy value (635.323×10^{-3} J), showing insufficient deformation of the pore walls.

Figure 8(d) shows that during cycle 3, specimens 3# and 2# demonstrate the highest ERR, at 51.438% and 54.251%, respectively. Specimen 3# undergoes collaborative deformation of struts and cell walls. Despite plastic buckling dominance, 2# demonstrates improved elastic recovery after densification. Figure 5(a) shows that 1# has the lowest ERR (46.482%), indicating poor elastic recovery and extensive local collapse.

As shown in Fig. 8(e), specimen 3# exhibits the lowest RDR in cycle 3, at less than 13%, indicating excellent reversibility due to its gradient structure and beam reinforcement. Specimen 2# follows closely behind 3# with

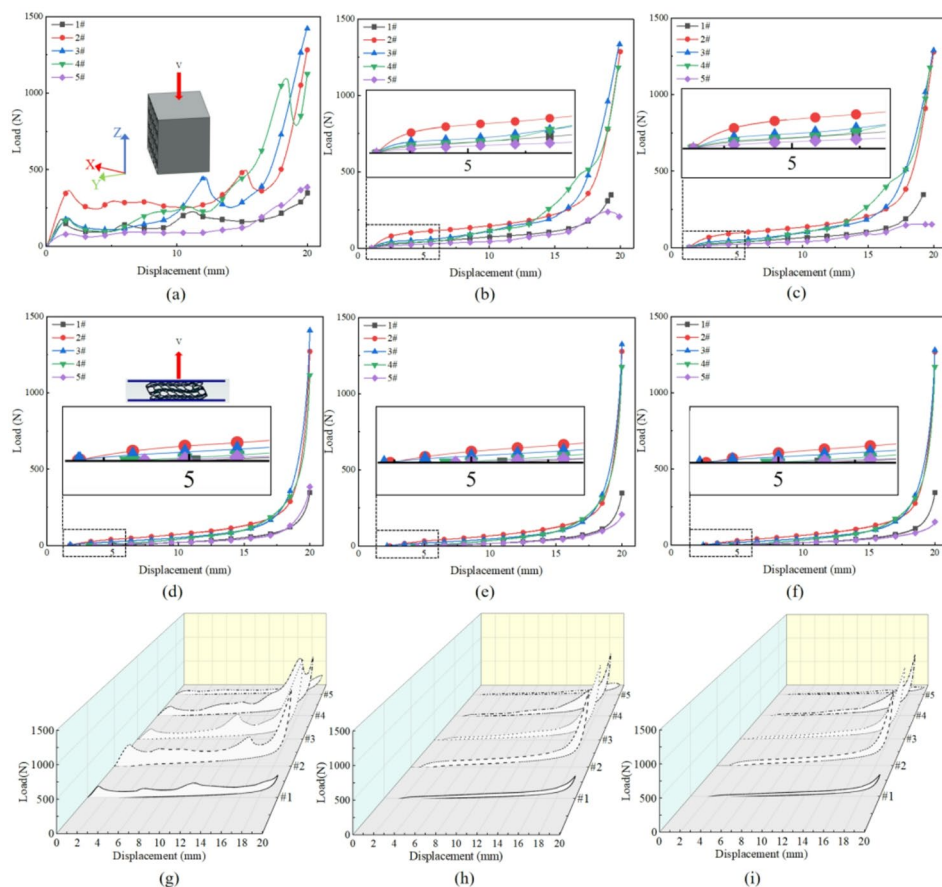


Fig. 7. Load–displacement curves of specimens 1# to 5# under three quasi-static compression cycles: (a) first loading, (b) second loading, (c) third loading, (d) first unloading, (e) second unloading, (f) third unloading, (g) first hysteresis loop, (h) second loop, (i) third loop.

Specimen	Cycle	EA ($\times 10^{-3}$ J)	SEA ($\times 10^{-3}$ J/g)	HLA ($\text{J} \times 10^{-3}$)	ERR (%)	RDR (%)	SDR (%)
1#	1	3142.801	246.494	2445.521	24.347	26.460	111.254
1#	2	1611.929	126.426	948.985	43.832	30.031	
1#	3	1454.382	114.069	819.913	46.482	31.726	
2#	1	7151.256	428.990	4820.564	32.904	16.635	106.906
2#	2	4291.937	257.465	2055.067	52.020	11.847	
2#	3	3979.978	238.751	1816.491	54.251	13.218	
3#	1	6348.750	434.251	4130.583	35.147	16.635	97.408
3#	2	4196.950	287.069	2102.551	49.416	11.847	
3#	3	3868.761	264.621	1857.820	51.438	12.156	
4#	1	6808.638	431.199	4948.120	28.007	16.635	114.038
4#	2	4041.560	255.957	2208.880	45.669	10.839	
4#	3	3719.179	235.540	1904.815	48.055	23.114	
5#	1	2478.141	244.876	1735.581	31.366	19.164	43.165
5#	2	1423.614	140.673	849.366	42.078	20.859	
5#	3	1124.577	111.124	635.323	45.891	24.664	

Table 3. Single-indicator results.

13.218%. Specimen 5# shows the highest RDR (24.664%), consistent with severe irreversible crushing and poor recovery, as shown in Fig. 5.

In Fig. 8(f), specimen 3# achieves the most promising stiffness degradation ratio (SDR) of 97.408%, indicating strong stiffness retention and mechanical stability over three loading cycles. This behavior is consistent with the

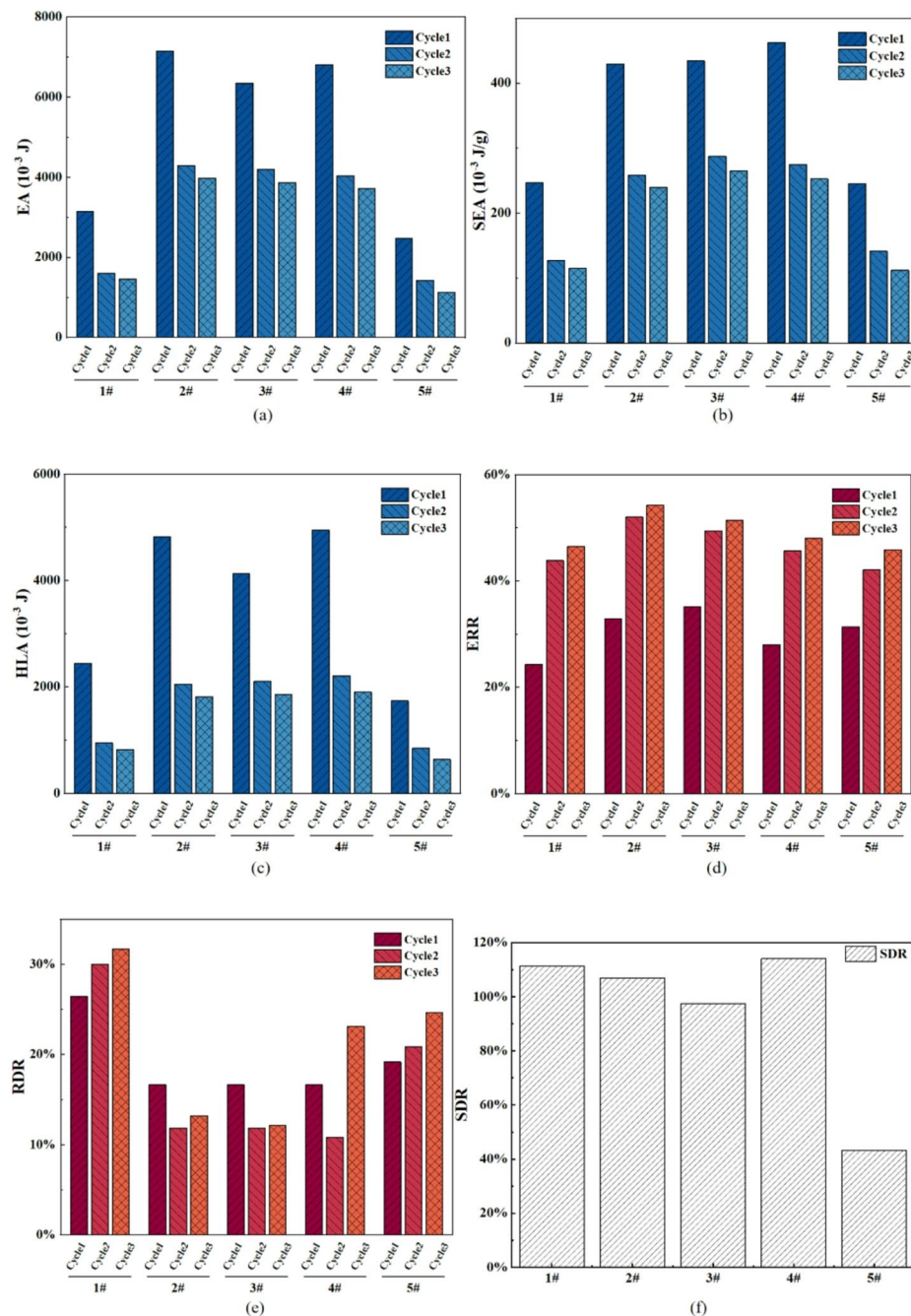


Fig. 8. Comparative bar charts of performance metrics across three cycles: (a) EA, (b) SEA, (c) HLA, (d) ERR, (e) RDR, and (f) SDR.

deformation mechanisms reported in⁴⁷, where polymer lattice structures exhibiting stable cyclic responses were shown to undergo gradual elastic–plastic transitions while preserving load-transfer pathways, thereby effectively delaying stiffness degradation. The gradient cell topology of specimen 3# promotes sequential participation of cells with different stiffness levels, leading to a more homogeneous deformation distribution and sustained macroscopic stiffness during repeated compression. In contrast, specimens 1#, 2#, and 4# exhibit SDRs exceeding 100% (e.g., specimen 4# reaches 114.037%), which does not reflect enhanced rebound performance but instead occurs due to the geometric locking caused by permanent deformation. This apparent stiffness increase leads to structural densification rather than genuine elastic recovery. From Figs. 8(d) and 8(e), it can be observed that although specimen 4# shows a high SDR, its energy recovery ratio (ERR) of 49.173% and residual displacement ratio (RDR) of 16.757% are both lower than those of specimen 3#, confirming its lower rebound efficiency. Similarly, specimens 1# and 2# exhibit elevated SDRs but suffer from lower ERRs and higher RDRs, further supporting the conclusion that a higher SDR does not necessarily indicate better rebound performance; instead, it may result from irreversible structural collapse. Specimen 5# shows the lowest SDR (43.165%), indicating

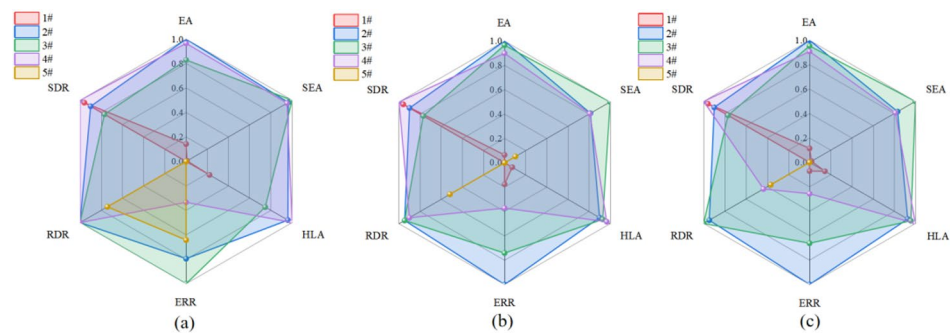


Fig. 9. Comprehensive performance evaluation model: (a) first cycle, (b) second cycle, and (c) third cycle.

Specimen	Cycle 1	Cycle 2	Cycle 3
1#	0.0014	0.0056	0.0066
2#	2.2735	2.2330	2.2715
3#	2.0455	2.0590	2.0350
4#	1.9510	1.7335	1.4080
5#	0.2065	0	0

Table 4. The radar chart area was calculated using Eq. (2.9) as a composite evaluation metric for energy absorption and rebound performance.

rapid stiffness degradation. Coupled with the highest RDR (24.661%) and a low ERR (47.598%), this indicates the weakest recovery capability, accompanied by fatigue and plastic damage under cyclic loading.

Comprehensive index results analysis

Figure 9 shows the performance of samples 1# to 5# across three loading cycles using six indicators: EA, SEA, HLA, ERR, RDR, and SDR. All indicators were normalized using Eqs. (7) and (8), which revealed the advantages and degradation trends of samples 1# to 5# during the cyclic process.

In Fig. 9, the radar profiles of specimens 2# and 3# are the largest and most complete, indicating that they maintained a high level of comprehensive performance across all three cycles. Among them, specimen 2# shows nearly overlapping SDR values in the first and third cycles that almost overlap, demonstrating the best performance stability.

Specimen 4# exhibits a clear attenuation trend. Although its radar profile in the first cycle remains close to those of specimens 2# and 3#, it gradually contracts in subsequent cycles, indicating a progressive decline in comprehensive performance as the number of cycles increases.

Specimens 1# and 5# display markedly contracted radar charts, reflecting overall weaker performance. In particular, specimen 5# has the lowest SDR value, with its radar profile nearly touching the center. This behavior indicates a lack of effective beam support, resulting in insufficient overall stiffness.

Overall, the radar profiles for Cycle 1 are generally larger than those for Cycles 2 and Cycle 3, indicating that all samples exhibit better comprehensive performance during the initial cycle, followed by gradual performance degradation in subsequent cycles.

The results in Table 4 show that specimens 2# and 3# have the largest radar-chart enclosed areas with only minor differences across the three cycles, indicating the highest and most stable overall performance. In particular, specimens 2# and 3# maintain radar-chart enclosed areas close to their peak values throughout all cycles, indicating a strong resistance to performance degradation under cyclic loading. This observation demonstrates that the six evaluated metrics EA, SEA, HLA, ERR, RDR, and SDR, for specimens 2# and 3# do not exhibit significant deterioration during repeated compression, suggesting good durability and resistance to mechanical fatigue⁴⁸.

By contrast, the area of specimen 4# gradually decreases with each cycle number, reflecting a marked decline in its comprehensive performance.

Specimen 1# and 5# maintain areas close to zero, consistent with their radar charts that nearly converge at the origin, indicating that their overall performance remains consistently low over time.

Overall, this study confirms that mechanical performance varies significantly with changes in topological configuration.

Conclusions

Five TPU specimens (1# to 5#) were fabricated with varying cell structural parameters using the FDM. Their energy absorption and resilience properties were evaluated through quasi-static compression tests. The main conclusions are summarized as follows:

- (1) In terms of energy absorption performance, the specimen 2# with homogeneous small-pores demonstrated the highest energy absorption (EA), reaching 3979.978 J in cycle 3. This occurs due to its dense pore distribution and short cell walls, facilitating the formation of extensive plastic buckling zones during compression and efficient energy dissipation. The specimen 3# gradient specimen exhibited the best performance in SEA, achieving 264.621 J/g in cycle 3, effectively balancing structural lightweighting with high energy dissipation capacity. In terms of rebound performance, the specimen 3# outperformed the others with an ERR of 49.416% and the lowest RDR of 11.847% in cycle 3, indicating excellent elastic recovery and reversible deformation characteristics.
- (2) In terms of comprehensive energy absorption and rebound performance, specimen 3# demonstrated the best overall performance. It outperformed other structures in SEA, ERR, and RDR, achieving a SEA of 264.621 J/g, an ERR of 51.44%, and an RDR of only 12.16% in cycle 3. It also exhibited robust stiffness retention, with an SDR of 97.41%. The radar chart showed a comprehensive performance area of 2.0350, comparable to that of specimen 2# (2.2715). However, specimen 3# demonstrated stable performance, indicating better synergy between energy absorption and rebound under repeated cyclical loading. The comprehensive evaluation proposed in this study overcomes the limitations of single-indicator assessments and provides a novel quantitative tool for the systematic design of dual-optimized structures that consider both energy absorption and rebound. These findings support the long-life structural design of UAVs operating in extreme service environments and have significant engineering value for enhancing the safety and stability of their vibration mitigation systems.

In summary, lattice geometry plays a critical role in influencing energy absorption and rebound performance. Smaller cell apertures and denser arrangements promote effective energy dissipation and dispersion during compression. Gradient and centrally symmetric structures promote hierarchical stress transfer, thereby enhancing energy absorption and controllability of deformation. Additionally, the incorporation of beam structures as interlayer supports significantly improves compressive stability and rebound capability by effectively transmitting elastic strain during unloading, which aids in pore recovery and stiffness retention. In contrast, structures without beams (e.g., specimen 5#) are prone to irregular collapse and plastic deformation, rendering them unsuitable for applications that require energy absorption and rebound under high-frequency loading conditions.

Data availability

The datasets generated and/or analysed during the current study are not publicly available due to the fact that the data are all obtained from the experiments mentioned in this paper but are available from the corresponding author on reasonable request.

Received: 10 October 2025; Accepted: 12 January 2026

Published online: 16 March 2026

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Funding

The work was supported by the Yunnan Fundamental Research Projects (Grant No. 202301AT070398) and the NSFC Project (Grant No. 62461160259).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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